

MD NAHID HASAN

SOFTWARE ENGINEER • AI/ML RESEARCHER

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFILE

Software Engineer and Computer Science PhD student with 2+ years of production experience at Samsung R&D Institute building Android/WearOS features for 10M+ users. Experienced in Kotlin, Java, system-level Android development, testing, debugging, CI/CD, and AI/LLM-oriented software engineering research.

EXPERIENCE

Graduate Teaching Assistant

2026 – Present

University of North Carolina at Charlotte, Charlotte, NC

- TA for ITIS 3300 – Software Requirements, Design, and Testing; assist students with requirements engineering, software design, testing, and project development.
- Researching applications of AI and Large Language Models (LLMs) to improve software engineering practices under Dr. Rezarta Krasniqi.

Machine Learning / Computer Vision Intern

Summer 2026

Innovision LLC

Remote

- Built a browser-based application for reviewing and correcting SAM2-predicted burn-wound segmentation polygons and visualizing segmentation overlays, alignment/feature JSON, and 3D .ply reconstructions.
- Deployed the application on Render with a Google Drive-backed data pipeline and GitHub-based development workflow.

Graduate Teaching Assistant

Aug 2024 – May 2026

Miami University, Oxford, OH

- Instructed and mentored students in Machine Learning, Artificial Intelligence, and Systems Engineering courses, supporting programming assignments and projects.

Software Engineer → Senior Software Engineer

Jul 2022 – Jul 2024

Samsung R&D Institute, Dhaka, Bangladesh

- Developed and shipped production Android features for the Galaxy Wearable ecosystem using Kotlin, Java, Android SDK, and WearOS APIs, supporting Galaxy Watch 6/7 and 10M+ active users worldwide.
- Designed system-level modules including Settings, Display, Sound, and Device Info using MVVM, ViewModel, and LiveData, improving modularity, maintainability, and runtime performance.
- Expanded unit and integration testing with JUnit and Mockito; diagnosed memory leaks, crashes, ANRs, and performance issues to reduce production regressions.
- Debugged cross-layer issues involving Bluetooth connectivity, framework APIs, firmware, and device compatibility in collaboration with QA and product teams.
- Contributed to CI-driven Gradle build pipelines, Git-based code reviews, release workflows, and performance profiling.

SELECTED PROJECTS

QuikDel – Hierarchical Multi-Agent Delivery System

Python, Q-Learning, Simulation

- Developed a hierarchical multi-agent Q-Learning system for urban delivery, reducing travel distance by up to 25% while handling 13,866+ delivery requests/hour in simulation.
- Published in *IEEE Access*, 2026; additional manuscript under review at ICAPS 2026.

Human Activity Recognition

Python, Scikit-learn, TensorFlow

- Built ML pipelines for sensor-based activity classification and compared Random Forest, XGBoost, CNN, and other models across 15 users and 15 activities.

EDUCATION

PhD in Computer Science

2026 – Present

University of North Carolina at Charlotte, Charlotte, NC

Research: Software Engineering for AI / LLMs

MSCS in Computer Science

Aug 2024 – May 2026

Miami University, Oxford, OH

GPA: 4.00/4.00

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C++, C, C#, SQL, JavaScript, HTML/CSS, Bash

Software: Android SDK, WearOS, Git, Linux, Docker, AWS, Gradle, MySQL, MongoDB, REST APIs, CI/CD

Engineering: OOP, DSA, MVC/MVVM, System Design, Unit Testing, Integration Testing, Debugging, Agile

AI/ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, LLMs, Reinforcement Learning, Computer Vision

ACHIEVEMENTS

Professional Level Certification in SW Certificate Test (2022) • Excellence Awards, Samsung R&D Institute (2023) • Codeforces Rating: 1544 • 1,300+ Problems Solved • ICPC Dhaka Regional Rank: 74th • Dean's List (3×), University of Rajshahi